Module 1 Project Rubric

RiskLens | Visual Defect Inspector

C1 - Visual Inspection and Evaluation
Build and evaluate a vision-language inspector that diagnoses and corrects the inherited baseline defects, returns reliable structured outputs, improves defect F1 on a frozen test set, and justifies prompting and image-resolution choices using measured quality and cost.

Evidence to prepare

- **D1.1:** Frozen test split of at least 40 images, with at least 20 good and 20 defective images; separate development set; printed seed and counts.
- **D1.2:** Three bug diagnoses, each showing a concrete symptom, root cause and implemented fix.
- **D1.3:** A working inspect(image) with the required verdict, defect_type and description fields, defensive parsing and a safe fallback; justified model choice.
- **D1.4:** At least three measured development-set prompt iterations, including a few-shot attempt, with the final choice explained.
- **D1.5:** Three analyzed test cases with images and hypotheses: use actual errors; if fewer than three exist, show all errors and enough borderline cases to reach three.
- **D1.6:** Saved Stage 1 evaluation: baseline and final defect F1, confusion matrix, timing and estimated cost per 1,000 inspections; final F1 must exceed baseline.
- **D1.7:** At least three image sizes compared on development accuracy and seconds per image, with a production-resolution recommendation.

How grading works

All 5 criteria must be **Met** to complete C1. **C1.1 is MUST PASS** and is reviewed first. If it is Not Yet, revise before the remaining criteria are reviewed. No points, weights or bonus credit.

Submit the same notebook link with saved outputs, a milestone snapshot, a short summary and known issues. Revise Not Yet criteria and affected evidence. The full RiskLens Grading Rubric defines review rules; course policy governs attempts and timing.

Module 1 | Assessment criteria (1/2)

C1 - Visual Inspection and Evaluation

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.1 Your evaluation split is frozen **MUST PASS**	D1.1 SPLIT cell; saved seed/counts; pre-iteration notebook snapshot or version history.	Before prompt iteration, freeze at least 40 test images with at least 20 per class and a separate development set. Print seed, membership identifiers and class counts. Preserve a dated snapshot or version record. Use development data for choices and disclose any later test exposure.	Changing the split after seeing results; overlapping development and test images; treating cell position alone as proof of timing.
C1.2 Your inspector fixes the inherited defects	D1.2 bug diagnoses; D1.3 inspect(); representative outputs and fallback checks.	Diagnose and fix the negation-unsafe parser, context-free prompt and empty defect description. Each diagnosis links a real symptom to its cause and fix. inspect() returns the required JSON fields and valid verdict, uses a domain prompt and structured output, and handles malformed model replies without an uncaught parsing error. Justify the VLM choice.	Leaving one bug unresolved; treating "no defects" as defective; returning empty defect evidence; a parser failure that crashes the pipeline.
C1.3 Your prompt iteration is measured	D1.3-D1.4 prompt ladder; saved development outputs.	Compare at least three prompt iterations on the same development set, including a few-shot attempt. Keep every prompt, development accuracy, reason for the change and final selection visible. Include the few-shot token-cost observation. Select the final prompt before final-test scoring.	Only showing the winning prompt; unmeasured prompt changes; choosing a prompt from test-set results; omitting the few-shot attempt.

Evidence note. Assess saved outputs and linked artifacts; graders do not need to repeat model training. A missing evaluation output does not demonstrate a result. Preserve real prompts, configurations, data versions and outputs as the work happens. Dated snapshots or version history support freezing; cell position alone does not prove chronology.

Module 1 | Assessment criteria (2/2)

C1 - Visual Inspection and Evaluation

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.4 Your performance and error claims are supported	D1.5 images and analysis; D1.6 saved Stage 1 evaluation and ledger.	On the frozen test set, report baseline and final defect F1 and the confusion matrix; final F1 exceeds inspect_v0. Analyze at least three actual errors with plausible hypotheses. If fewer exist, analyze all errors and enough clearly labeled borderline cases to total three. Save the evaluation outputs and explain the baseline recall inflation.	Comparing different test sets; substituting accuracy or recall for defect F1; inventing errors; reporting numbers without saved outputs.
C1.5 Your resource recommendation follows the evidence	D1.6 timing/cost; D1.7 resolution table and recommendation.	Measure development accuracy and seconds per image at three or more image sizes under stated hardware and settings. Record estimated cost per 1,000 inspections with the assumed hourly price. Recommend an image size based on the observed quality/time trade-off and explain what the processor actually does with the resized inputs.	Only reporting memory; comparing different image sets; claiming higher input resolution always helps; omitting price or runtime assumptions.

Module handoff. Freeze the split before experimentation. If reassessment exposes the test set, record that exposure and agree a new evaluation version before further tuning. Do not manufacture three errors when fewer exist.

Source mapping: D1.1-D1.7. RiskLens Criterion Set v2.0 governs criteria; the published course policy governs administrative rules.